Kaustubh Gupta

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RESEARCH INTERESTS

Multilingual trustworthy AI; task-conditioned over-refusal in aligned LLMs; mechanistic interpretability; agency-preserving alignment

EDUCATION

BTech in Artificial Intelligence and Engineering Science - 9.34 Hyderabad, IN | 2025
Indian Institute of Technology Hyderabad [Highest CGPA in Department]
Teaching Assistant: Convex Optimization (Jan-May 2024, Jan-May 2025)
Coursework: Probability, Convex Optimization, ML, Deep Learning, RL, NLP, Speech Systems, AI For Humanity

COURSES

BETTER FUTURES FELLOWSHIP | Participant Remote | Aug 2026 – Nov 2026

- AI Safety Collab pilot on Forethought Better Futures curriculum: post-AGI futures, cause prioritization, and longtermist framing alongside technical safety work.

WORK EXPERIENCE

ORACLE | Data Scientist 1 Bangalore, IN | Jul 2025 – Present

- Ship LLM query-understanding for Fusion Analytics Warehouse Copilot: entity classification, column reranking, and retrieval-backed generation.
- Improve Copilot quality via prompt tuning, JSON response contracts, and production backend-log analysis.

SENTIENT FUTURES | Fellow Remote | Aug 2026 – Nov 2026

- Mentored research with John Lund (AE Studio) on robustness of moral consideration under adversarial pressure in language models.

ORACLE | Data Analytics Intern Bangalore, IN | May 2024 – Jul 2024

- Built LLM synthetic-data pipeline for Oracle FAW (Python, K-Means/farthest-point sampling); adopted by a Consulting MTS.

PUBLICATIONS

Interpreting and Steering State-Space Models via Activation Subspace Bottlenecks | ICML 2026
Co-author; mechanistic interpretability on Mamba LLMs via activation subspace bottlenecks; benchmarked steering controllability vs. quality.

Stream-TTS: A Low-Latency Text-to-Speech using Kolmogorov-Arnold Networks | IEEE ICASSP 2025
Co-authored multilingual streaming TTS with KANs; 0.0795 RTF and 4.09 MOS with prosody control.

Fairness-Aware Submodular Optimization: A Survey | Unpublished survey, IIT Hyderabad
Sole-authored survey of fairness constraints in submodular maximization (summarization, influence spread); synthesized fairness–utility tradeoffs and open directions for equitable ML selection.

Attentive Multimodal Learning for Suicide Risk Assessment Using Acoustic and Prosodic Cues | Under review, Speech Communication
Built initial multimodal pipeline with acoustic embeddings and prosodic cues for suicide risk classification.

PROJECTS

PREFKIT - MULTI-METHOD PREFERENCE ELICITATION TOOLKIT FOR OPEN WEIGHT MODELS

Python, PyTorch, HuggingFace

Implemented a toolkit containing four preference elicitation methods (pairwise, Likert, action-choice, best-worst scaling). Scored these methods on Qwen3 1.7B–14B on a 24-outcome set. Concluded that one-method surveys are not sufficient to prove a model's preferences, and cross-method agreement is not monotone with size.

MULTILINGUAL LLM OVER-REFUSAL EVALUATION

Python, PyTorch, HuggingFace, OR-Bench

Replicated Sakonii task-over-refusal benchmark on Llama-3.1-8B with calibrated OR-Bench-style judge; extending evaluation to Hindi/Hinglish task wrappers with human-reviewed translations.

OPTICAL CHARACTER RECOGNITION FOR ANCIENT SANSKRIT MANUSCRIPTS

Python, OCR, CRNN

OCR for degraded Sanskrit manuscripts via line segmentation, CRNN fine-tuning, and multimodal LLM evaluation.

AI-POWERED SESSION HIJACKING PREVENTION

Python, XGBoost, Flask, Redis

HTTP anomaly detector on CSIC 2010 (F1 0.97, SMOTE) with Redis-backed Flask inference and admin dashboard.

SUDOKU WEB APPLICATION

React, Django, Python

Full-stack Sudoku game with puzzle generation, backtracking solver, and interactive UI.

SPOTIDB

Python, Spotify API, MySQL

Spotify-inspired music database modeling tracks, artists, albums, playlists, and user ratings via the Spotify API.

DATA STRUCTURES & ALGORITHMS FROM SCRATCH

C++

Core data structures and algorithms implemented from first principles with emphasis on memory management and complexity trade-offs.

RESEARCH ARTIFACTS

TINY-AYA GLOBAL BLINDSPOTS

Python, HuggingFace, LLM Probing

Curated failure-case dataset (26/259) for CohereLabs/tiny-aya-global (3.35B); red-team probe across safety, bias, reasoning, and security scenarios.

ALLM FAIRNESS PROMPTS

Python, HuggingFace, LLM Eval

Fairness evaluation prompt set for aligned language models; used to assess bias and stereotyping in multilingual LLM outputs.

SKILLS

Languages: Python, C++, C, SQL, JavaScript, Bash
Technology: PyTorch, HuggingFace, LLM Eval, Prompt Engineering, Retrieval/Reranking, XGBoost, Flask, Redis, Git, Docker, LaTeX